

Ian Petrow: Curriculum Vitae

CONTACT INFORMATION	University College London Department of Mathematics 25 Gordon Street London WC1H 0AY United Kingdom		<i>website:</i> ianpetrow.com <i>Tel:</i> +44 (0)20 7679 8097 <i>E-mail:</i> i.petrow@ucl.ac.uk
RESEARCH INTERESTS	Number theory, especially the analytic theory of automorphic forms and L -functions		
EMPLOYMENT	University College London University College London ETH Zurich EPF Lausanne	Associate Professor Lecturer (equiv. Assistant Prof) Ambizione Fellow Post-Doc	Oct 2021 – Present Oct 2019 – Sept 2021 Feb 2017 – Oct 2019 Aug 2013 – Jan 2017
EDUCATION	Stanford University University of Cambridge Princeton University	Ph.D. M.A.St. A.B.	Sep 2013 Jun 2008 Jun 2007
GRANTS AND AWARDS	Faculty Education Award for Individual Excellence Frontiers of Science Award, ICBS Beijing EPSRC New Investigator Award Ambizione Fellowship, Swiss NSF AMS-Simons Travel Grant Cambridge Overseas Trust Bursary		2024 2024 2022-2025 2017-2020 2014-2016 2007-2008
RESEARCH PUBLICATIONS	1. The cubic moment of L-functions for specified local component families, (with Y. Hu and M.P. Young). <i>J. Eur. Math. Soc. (JEMS)</i>. To appear, (2026). 2. A generalized PGL(2) Petersson/Bruggeman-Kuznetsov formula for analytic applications, (with Y. Hu and M.P. Young). <i>Forum Math. Sigma</i>. 14 Paper No. e27, 113 pp. (2026). 3. The Weyl Law for Algebraic Tori. <i>J. Eur. Math. Soc. (JEMS)</i>. 26 no. 7, pp. 2441-2532 (2024). 4. The Fourth Moment of Dirichlet L-functions Along a Coset and the Weyl Bound, (with M.P. Young). <i>Duke Math. J.</i> 172 no. 10, pp. 1879-1960 (2023). 5. The Volumes of Miyauchi Subgroups, (with D. Lesesvre). <i>Arch. Math. (Basel)</i>. 119, pp. 37-51 (2022). 6. The Weyl Bound for Dirichlet L-functions of Cube-Free Conductor, (with M.P. Young). <i>Ann. of Math. (2)</i>. 192 no. 2, pp. 437-486 (2020). 7. Oscillatory Integrals with Uniformity in Parameters, (with E.M. Kiral and M.P. Young). <i>J. Théor. Nombres Bordeaux</i>. 31 no. 1, pp. 145-159 (2019).		

8. Bounds for Traces of Hecke Operators and Applications to Modular and Elliptic Curves Over a Finite Field.
Algebra Number Theory. **12** no. 10, pp. 2471-2498 (2018).
9. A Generalized Cubic Moment and the Petersson Formula for Newforms, (with M.P. Young).
Math. Annalen. **373** no. 1-2, pp. 287-353 (2019).
10. Counting Automorphic Forms on Norm One Tori, (with E.H. Brooks).
Acta Arithmetica. **183** no. 2, pp. 117-143 (2018).
11. Elliptic Curves Over a Finite Field and the Trace Formula, (with N. Kaplan).
Proc. Lond. Math. Soc. **115** no. 6, pp. 1317-1372 (2017).
12. Traces of Hecke Operators and Refined Weight Enumerators of Reed-Solomon Codes, (with N. Kaplan).
Trans. Amer. Math. Soc. **370** no. 4, pp. 2537-2561 (2018).
13. A Twisted Motohashi Formula and Weyl-Subconvexity for L -functions of Weight Two Cusp Forms.
Math. Annalen. **363** no. 1-2, pp. 175-216 (2015).
14. A Recursion Formula for Moments of Derivatives of Random Matrix Polynomials, (with A. Altuğ, S. Bettin, Rishikesh, and I. Whitehead).
Quarterly J. Math. **65** no. 4, pp. 1111-1125 (2014).
15. Moments of $L'(1/2)$ in the Family of Quadratic Twists.
Int. Math. Res. Notices. **2014** no. 6, pp. 1576-1612 (2014).
16. Transition Mean Values of Shifted Convolution Sums.
J. Number Theory. **133** no. 10, pp. 3264-3282 (2013).

THESES, ESSAYS,
AND REPORTS

1. Cubic Moments and a Distance Function on Cuspidal Representations, (with M.P. Young).
in Automorphic Forms and Arithmetic, *Oberwolfach Reports*. no. 26, pp. 31-34 (2020).
2. Moments of Automorphic L -functions and Related Problems.
Ph.D. Thesis. Stanford University (2013).
3. Vinogradov's Three Primes Theorem.
Part III Essay. University of Cambridge (2008).
4. Numerical Evidence for a Conjecture of Hooley.
Undergraduate Thesis. Princeton University (2007).

LONG-TERM

RESEARCH VISITS

Institut Mittag-Leffler, Djursholm, Sweden
 Institut Mittag-Leffler, Djursholm, Sweden (cancelled; moved online)
 Forschungsinstitut für Mathematik, Zürich, Switzerland (cancelled)

Apr 2024

Jan-Feb 2021

Jun-Aug 2020

SELECTED LECTURES	Keynote Speaker, Dynamics, Equations and Applications, Krakow, Poland	Jun 2027
	LMS-Sheffield Symposium on The Langlands Programme, Sheffield, UK	Jul 2026
	Group Actions and Harmonic Analysis in Number Th., Oberwolfach, Germany	May 2026
	CMI workshop Zeta and L -functions, Oxford, UK	Sep 2025
	Analytic Number Theory and Automorphic Forms, Lille, France	Sep 2025
	ICBS Frontiers of Science in Mathematics Award Lecture, Beijing, China	Jul 2024
	Mini-course, Hausdorff Center for Mathematics Summer School, Bonn, Germany	Jul 2024
	CMI workshop New Perspectives in Analytic Th. of Aut. Forms, Oxford, UK	Sep 2023
	Automorphic Forms and L -functions of higher rank, London, UK	Sep 2023
	Oxford Number Theory Seminar, Oxford, UK	Nov 2022
	Automorphic Forms Conference, Erdős Center, Budapest, Hungary	Sep 2022
	L -functions, Circle-Method and Applications, ICTS Bengaluru (via Zoom)	Jun 2022
	Hausdorff Institute Seminar: Harmonic Analysis from the Edge, (via Zoom)	Aug 2021
	Automorphic Forms and Arithmetic, Oberwolfach, Germany	Sep 2020
	Automorphic Forms in Budapest 2020, (via Zoom)	Jun 2020
	Second Symposium on Analytic Number Theory, Cetraro, Italy	July 2019
	Séminaire de Théorie des Nombres de l'IMJ-PRG, Paris, France	Jan 2019
	Workshop on Geometric and Analytic Number Theory, Göttingen, Germany	Nov 2018
	Joint IAS/Princeton University Number Theory Seminar, Princeton, NJ	Mar 2018
	Mini-course, MSRI Introductory Workshop, Berkeley, CA	Feb 2017
	AMS Special Session, Joint Mathematics Meetings, Atlanta, GA	Jan 2017
	MAA Invited Paper Session, Joint Mathematics Meetings, Atlanta, GA	Jan 2017
	Spectral Theory, Automorphic Forms and Arithmetic, Copenhagen, Denmark	Nov 2016
	Geometric and Analytic Number Theory, Zürich, Switzerland	Sep 2016
	AMS Special Section on Modern Methods in Analytic NT, Oxford, MS	Mar 2013
	Hawai'i Conference in Algebraic Number Theory, Arithmetic Geometry and Modular Forms, Honolulu, HI	Mar 2012

TEACHING
EXPERIENCE

Department of Mathematics, University College London

Lecturer

Prime Numbers and their Distribution	Fall 2025
Prime Numbers and their Distribution	Fall 2024
Prime Numbers and their Distribution	Fall 2023
Prime Numbers and their Distribution	Fall 2022
Prime Numbers and their Distribution	Fall 2021
Prime Numbers and their Distribution	Fall 2020
Prime Numbers and their Distribution	Spring 2020

Post-doc supervised:

Chung-Hang Kwan	EPSRC Research Fellow	2022-2025
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Students supervised:

Matteo Di Scipio	PhD	2024-
Alberto Acosta Reche	PhD	2023-
Catinca Mijdei	PhD	2023-
Yinghao Wang	4th Year MSci Project	2024-2025
Seth Hardy	4th Year MSci Project	2021-2022
Sarah Al-Ali	4th Year MSci Project	2020-2021
Travor Liu	LMS-funded summer research	2024
Charlotte Clare-Hunt	LMS-funded summer research	2023
Jieun Kim	Dept.-funded summer research	2022

Department of Mathematics, ETH Zürich

Lecturer

Introduction to Analytic Number Theory	Fall 2018
Classical Modular Forms	Fall 2017

Students supervised:

Fabian Lenz	Bachelor's Thesis	2019
Milos Radosavljevic	Bachelor's Thesis	2018
Felix Schmid	Master's Thesis	2017-2018

Department of Mathematics, École Polytechnique Fédérale de Lausanne

Lecturer

Analytic Theory of L -Functions (Ph.D. course)	2015-2016
Introduction to Analytic Number Theory	

Senior teaching assistant

Introduction to Algebraic Number Theory	2013-2016
Galois Theory	
Introduction to Analytic Number Theory	
Linear Algebra for Computer Scientists	

Students supervised:

Gauthier Leterrier	Master's Semester Project	Fall 2016
Sébastien Comtesse	Bachelor's Semester Project	Spring 2016
Joachim Petit	Master's Semester Project	Fall 2015
Raphaël Zacharias	Master's Semester Project	Fall 2014
Alice Hedenlund	Bachelor's Semester Project	Spring 2014
Ilqar Ramazanli	Master's Semester Project	Fall 2013

Canada/USA Mathcamp

Teaching mentor with tenure

Taught and mentored talented high school students undergraduate and graduate level mathematics in an intensive summer program.	Summer 2011
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Department of Mathematics, Stanford University

Teaching assistant

2009-2013

Differential Equations (twice)

Honors Multivariable Calculus

Single Variable Calculus I

Single Variable Calculus II

Champlain Valley Union High School, VT

Long-Term Substitute Teacher

Spring 2005

Various High School Math Courses

QUALIFICATIONS

Fellow of the Higher Education Academy, United Kingdom

Nov 2021

Qualified for associate professor of mathematics in Denmark

Feb 2019

Qualifié pour le corps des professeurs des universités en France

Feb 2019

EDITORIAL

Editor of *Mathematika*

2023 -

SERVICE

Departmental Positions:

Hiring Committee Member (multiple positions)

2024-

4th Year MSci Projects Coordinator

2021-2024

Affiliate Tutor (Study Abroad and Liaison)

2020-2023

2nd Year Group Projects Alg./NT/Combinatorics stream leader

2020-2022

Seminar organization:

London Learning Seminar on Analytic Number Theory

2020-

Egham, Reading, London Arithmetic Statistics Seminar (ERLASS)

2020-2021

EPFL Number Theory Seminar

2015-2016

Referee: > 30 full reports for a wide variety of journals

Reviewer: Math Reviews; Zentralblatt Math

Memberships: AMS, LMS

**CONFERENCE
ORGANIZATION**

“Explicit methods in automorphic forms”, UCL Department of Mathematics, 19-20 June 2025. Organized jointly with Chung-Hang Kwan. Funded by EPSRC.

CAREER BREAKS

Paternity Leave - Jan 13, 2020 to Feb 7, 2020

Paternity Leave - Oct 23, 2023 to Nov 17, 2023

LANGUAGES

English (Native), French (Fluent), German (B1).